deduction version 2

Below is a rigorous derivation of the REM posterior odds, considering an activation filter and specific origins for targets and confusing foils. We will first assume a uniform non-match likelihood $f_N(s_i)$ and then show how it adapts for type-specific non-match likelihoods $f_{N,type(i)}(s_i)$.

Glossary of Terms (Applicable to Both Sections)

Symbol	Meaning	
M	Total number of **activated** traces after any first-stage filter. The similarity vector D (or $\mathbf{s}$) has M components.	
n	Number of items in the **current study list**. All n are assumed to be activated. Let A be the set of indices for these n traces.	
n'_{phys}	*Physical* number of items in the **most-recent prior list** from which confusing foils (CF) are sampled. Let B_{phys} be this set of indices.	
$n^\dagger$	Number of **activated** traces from the set B_{phys}. Let B_{act} be this set of $n^\dagger$ activated traces ($B_{act} \subseteq B_{phys}$).	
x	Prior probability that a foil trial is a confusing foil (CF).	
$1-x$	Prior probability that a foil trial is a pure new foil (NF).	
D	The observed data, i.e., the vector of M similarity scores $(s_1, s_2, \ldots, s_M)$ from the activated traces.	
O	Event that the test item is an "Old" target (from the current list).	
N	Event that the test item is a "New" foil (either CF or NF).	
S_j	Event that trace j is the true source/match for the test item.	
N_j	Event that trace j is a non-match for the test item.	
$f_M(s_j)$	Likelihood $P(D_j \mid S_j)$ of observing data s_j from trace j if it's a match.	

Section 1: Derivation Assuming a Uniform $f_N(s_i)$

In this section, we assume a single likelihood function $f_N(s_i) = P(D_i|N_i)$ for all non-matching components i.
Let $\lambda_k = \frac{f_M(s_k)}{f_N(s_k)}$.

Rigorous Derivation Steps (Uniform f_N)

The goal is to compute the posterior odds $\frac{P(O|D)}{P(N|D)}$.

$$\begin{aligned}
\frac{P(O \mid D)}{P(N \mid D)} &= \frac{P(D \mid O)P(O)}{P(D \mid N)P(N)} && \text{(Bayes' Theorem)} \\
&= \frac{P(D \mid O)}{P(D \mid N)} && \text{(Assuming equal priors } P(O) = P(N)\text{)}
\end{aligned}$$

1.1. Numerator: Likelihood $P(D \mid O)$ (Uniform f_N)

Targets (O) are drawn only from the n current list items (set A, all assumed activated). Given O, each of these n items is equally likely to be the source.

$$\begin{aligned}
P(D \mid O) &= \sum_{j \in A} P(D \mid S_j \text{ and } O)P(S_j \mid O) && \text{(Law of Total Probability over sources } j \in A\text{)} \\
&= \sum_{j \in A} P(D \mid S_j)\frac{1}{n} && \text{(Since } P(S_j|O) = 1/n \text{ for } j \in A\text{, and 0 otherwise)} \\
&= \frac{1}{n} \sum_{j \in A} \left[f_M(s_j) \prod_{\substack{i=1 \\ i \neq j}}^{M} f_N(s_i) \right] && \text{(If } j \text{ is match, } D_j \sim f_M,\ \text{other } D_i \sim f_N \text{ over } M \text{ activat}
\end{aligned}$$

1.2. Denominator: Likelihood $P(D \mid N)$ (Uniform f_N)

A foil (N) is either a confusing foil (CF) with probability x, or a new foil (NF) with probability $1 - x$.

$$P(D \mid N) = xP(D \mid CF) + (1 - x)P(D \mid NF)$$

- **1.2a. Likelihood $P(D \mid CF)$ (Confusing Foil, Uniform f_N):**
  Confusing foils are sampled uniformly from the n'_{phys} items in the physical prior list (B_{phys}).
  If the sampled item $j \in B_{phys}$ is activated ($j \in B_{act}$), its $s_j \sim f_M$ and other $M - 1$ activated $s_i \sim f_N$.
  If the sampled item $j \in B_{phys}$ is *not* activated ($j \notin B_{act}$), then for the M *activated* traces, none are a match, so $D \sim \prod_{i=1}^{M} f_N(s_i)$.

$$
\begin{aligned}
P(D \mid CF) &= \sum_{j\in B_{phys}} P(D \mid S_j \text{ and } CF)P(S_j \mid CF) && \text{(LTP over sources} \\
&= \frac{1}{n'_{phys}} \sum_{j\in B_{phys}} P(D \mid S_j) && \text{(Since } P(S_j|CF) = \\
&= \frac{1}{n'_{phys}} \left(\sum_{j\in B_{act}} \left[f_M(s_j) \prod_{\substack{i=1\\ i\neq j}}^{M} f_N(s_i) \right] + \sum_{j\in B_{phys}\setminus B_{act}} \left[\prod_{i=1}^{M} f_N(s_i) \right] \right) \\
&= \frac{1}{n'_{phys}} \left(\sum_{j\in B_{act}} \left[f_M(s_j) \prod_{\substack{i=1\\ i\neq j}}^{M} f_N(s_i) \right] + (n'_{phys} - n^\dagger) \prod_{i=1}^{M} f_N(s_i) \right)
\end{aligned}
$$

- **1.2b. Likelihood $P(D \mid NF)$ (New Foil, Uniform f_N):**
  A pure new foil matches none of the M activated traces.

$$
P(D \mid NF) = \prod_{i=1}^{M} f_N(s_i)
$$

1.3. Combine and Cancel Common Product $P_0(D) = \prod_{i=1}^{M} f_N(s_i)$ (Uniform f_N)

Recall $\lambda_k = f_M(s_k)/f_N(s_k)$. Any term $f_M(s_k)\prod_{i\neq k} f_N(s_i)$ can be written as $\lambda_k P_0(D)$.

- $P(D \mid O) = \frac{1}{n}\sum_{j\in A}\lambda_j P_0(D)$
- $P(D \mid CF) = \frac{1}{n'_{phys}}\left(\sum_{j\in B_{act}}\lambda_j P_0(D) + (n'_{phys} - n^\dagger)P_0(D)\right)$
- $P(D \mid NF) = P_0(D)$

Substituting into $\frac{P(D|O)}{xP(D|CF)+(1-x)P(D|NF)}$:

$$
\begin{aligned}
\frac{P(O \mid D)}{P(N \mid D)} &= \frac{\frac{1}{n}\sum_{j\in A}\lambda_j P_0(D)}{x\left[\frac{1}{n'_{phys}}\left(\sum_{j\in B_{act}}\lambda_j P_0(D) + (n'_{phys} - n^\dagger)P_0(D)\right)\right] + (1-x)P_0(D)} \\
&= \frac{\frac{1}{n}\sum_{j\in A}\lambda_j}{x\left[\frac{1}{n'_{phys}}\left(\sum_{j\in B_{act}}\lambda_j + (n'_{phys} - n^\dagger)\right)\right] + (1-x)} && \text{(Cancel } P_0(D) \text{ fro} \\
&= \frac{\frac{1}{n}\sum_{j\in A}\lambda_j}{x\left(\frac{1}{n'_{phys}}\sum_{j\in B_{act}}\lambda_j + \frac{n'_{phys}-n^\dagger}{n'_{phys}}\right) + (1-x)} && \text{(Distribute } 1/n'_{phy}
\end{aligned}
$$

Final Exact Odds Formula (Assuming Uniform f_N)

This is the rigorous expression for the posterior odds, given the activation filter, specific sources for targets and confusing foils, AND assuming a single $f_N(s_i)$ for all non-matching components.

$$\boxed{\frac{P(O \mid D)}{P(N \mid D)} = \frac{\frac{1}{n}\sum_{k \in A} \frac{f_M(s_k)}{f_N(s_k)}}{x\left(\frac{1}{n'_{phys}}\sum_{k \in B_{act}} \frac{f_M(s_k)}{f_N(s_k)} + \frac{n'_{phys} - n^\dagger}{n'_{phys}}\right) + (1-x)}}$$

- The sums $\sum_{k \in A}$ and $\sum_{k \in B_{act}}$ are over the respective sets of **activated** traces.
- The term $\frac{n'_{phys} - n^\dagger}{n'_{phys}}$ accounts for confusing foils sampled from physical prior list items that were *not* activated. If all n'_{phys} items are activated ($n^\dagger = n'_{phys}$), this term becomes zero.

Section 2: Derivation with Type-Specific $f_{N,type(i)}(s_i)$

You're absolutely right to scrutinize that step! It's crucial. Let's clarify exactly why $P_0^*(D)$ can still be defined and cancelled, even when the individual $f_{N,type(i)}(s_i)$ functions are different for each trace type.

Explanation (why can it be adapted)

1. Defining the "Baseline" Noise Product $P_0^*(D)$

When we introduce type-specific non-match likelihoods, we define $P_0^*(D)$ as the joint likelihood of observing the entire data vector $D = (s_1, s_2, \ldots, s_M)$ under the condition that **all M activated traces are non-matches of their respective specific types**.

So, $P_0^*(D) = \prod_{i=1}^{M} f_{N,type(i)}(s_i)$.

- This means $P_0^*(D) = f_{N,type(1)}(s_1) \times f_{N,type(2)}(s_2) \times \ldots \times f_{N,type(M)}(s_M)$.
- It's critical to understand that $P_0^*(D)$ is still a single, well-defined value for a given data vector $\mathbf{s}$, even if $f_{N,type(1)}(\cdot)$ is a different mathematical function than $f_{N,type(2)}(\cdot)$, and so on. It's simply the product of the likelihoods of each observed s_i under its own trace's specific non-match characteristic.
- This $P_0^*(D)$ correctly represents $P(D \mid NF)$ (the likelihood of the data given a Pure New Foil, where no activated trace is a match).
- It also correctly represents the likelihood of the data for a Confusing Foil that was sampled from a physical list item $j \in B_{phys}$ which was *not activated* ($j \notin B_{act}$). In this case, for the M

activated traces, none of them are the source of the CF, so they all behave as non-matches of their respective types.

2. How a "Match" Term Relates to $P_0^*(D)$

Now, consider a situation where one specific activated trace, say trace j, *is* the match (either as a target or an activated confusing foil). The likelihood of the data D in this case is:
$P(D \mid S_j) = f_M(s_j) \times \prod_{\substack{i=1 \\ i \neq j}}^{M} f_{N,type(i)}(s_i)$

We can algebraically rewrite this term by multiplying and dividing by $f_{N,type(j)}(s_j)$ (assuming $f_{N,type(j)}(s_j) \neq 0$, which is necessary for s_j to be observable under that non-match hypothesis):

$P(D \mid S_j) = f_M(s_j) \times \frac{\prod_{i=1}^{M} f_{N,type(i)}(s_i)}{f_{N,type(j)}(s_j)}$

$P(D \mid S_j) = \frac{f_M(s_j)}{f_{N,type(j)}(s_j)} \times \left(\prod_{i=1}^{M} f_{N,type(i)}(s_i)\right)$

Let $\lambda_j^* = \frac{f_M(s_j)}{f_{N,type(j)}(s_j)}$. This λ_j^* is the likelihood ratio for trace j, using its own specific non-match likelihood in the denominator.

Then, $P(D \mid S_j) = \lambda_j^* \times P_0^*(D)$.

3. Why This Still Allows Cancellation

Because *every* term in the expanded likelihoods can be expressed as some coefficient times this *same* $P_0^*(D)$:

- $P(D \mid O) = \left(\frac{1}{n} \sum_{j \in A} \lambda_j^*\right) P_0^*(D)$
- The part of $P(D \mid CF)$ from activated confusing foils: $\left(\frac{1}{n'_{phys}} \sum_{j \in B_{act}} \lambda_j^*\right) P_0^*(D)$
- The part of $P(D \mid CF)$ from non-activated confusing foils: $\left(\frac{n'_{phys} - n^\dagger}{n'_{phys}} \times 1\right) P_0^*(D)$ (because for these, the likelihood of D *is* $P_0^*(D)$)
- $P(D \mid NF) = 1 \times P_0^*(D)$

When you substitute these into the odds ratio:

$$\frac{P(O \mid D)}{P(N \mid D)} = \frac{\left(\frac{1}{n} \sum_{j \in A} \lambda_j^*\right) P_0^*(D)}{x\left[\left(\frac{1}{n'_{phys}} \sum_{j \in B_{act}} \lambda_j^* + \frac{n'_{phys} - n^\dagger}{n'_{phys}}\right) P_0^*(D)\right] + (1-x) P_0^*(D)}$$

The term $P_0^*(D) = \prod_{i=1}^{M} f_{N,type(i)}(s_i)$ is a common factor in the numerator and in every additive part of the denominator. Therefore, it **cancels out perfectly**.

What you might have missed:

The key is not that all $f_{N,type(i)}(s_i)$ have to be the *same function* for cancellation to occur. It's that the product $\prod_{i=1}^{M} f_{N,type(i)}(s_i)$ serves as the universal "baseline" likelihood when *no specific activated trace j is designated as the unique source/match*. When a specific trace j *is* the source, its individual $f_{N,type(j)}(s_j)$ contribution to this baseline product is replaced by $f_M(s_j)$. The ratio of $f_M(s_j)$ to $f_{N,type(j)}(s_j)$ (which is λ_j^*) is precisely what quantifies the change from that baseline due to the match at j.

So, the deduction shown in the image you screenshotted (which was my corrected rigorous derivation) remains algebraically sound even with type-specific f_N functions, provided that λ_k^* is understood to be $f_M(s_k)/f_{N,type(k)}(s_k)$. The complexity moves into defining and calculating each $f_{N,type(k)}(s_k)$, but the overall structure of the final odds ratio, after cancellation, is preserved.

Now The Deduction

Now, we address Steve's point by allowing the non-match likelihood $f_N(s_i)$ to be specific to the type of trace i, denoted $f_{N,type(i)}(s_i)$. This $f_{N,type(i)}(s_i)$ could itself be a mixture: $f_{N,type(i)}(s_i) = \sum_{k'} P(D_i = s_i \mid N_i, \text{subtype} = k')P(N_i \text{ is subtype } k')$.

Let $\lambda_k^* = \frac{f_M(s_k)}{f_{N,type(k)}(s_k)}$.

Rigorous Derivation Steps (Type-Specific f_N)

The overall structure of the odds $\frac{P(D|O)}{P(D|N)}$ remains the same. The divergence occurs in the expansion of the product terms.

2.1. Numerator: Likelihood $P(D \mid O)$ (Type-Specific f_N)

$$P(D \mid O) \;=\; \frac{1}{n} \sum_{j \in A} \left[f_M(s_j) \prod_{\substack{i=1 \\ i \neq j}}^{M} f_{N,type(i)}(s_i) \right]$$

2.2. Denominator: Likelihood $P(D \mid N)$ (Type-Specific f_N)

$$P(D \mid N) \;=\; xP(D \mid CF) \;+\; (1-x)P(D \mid NF)$$

- **2.2a. Likelihood $P(D \mid CF)$ (Confusing Foil, Type-Specific f_N):**

$$P(D \mid CF) \;=\; \frac{1}{n'_{phys}} \left(\sum_{j \in B_{act}} \left[f_M(s_j) \prod_{\substack{i=1 \\ i \neq j}}^{M} f_{N,type(i)}(s_i) \right] + (n'_{phys} - n^{\dagger}) \prod_{i=1}^{M} f_{N,type(i)}(s_i) \right)$$

- **2.2b. Likelihood $P(D \mid NF)$ (New Foil, Type-Specific f_N):**

$$P(D \mid NF) \;=\; \prod_{i=1}^{M} f_{N,type(i)}(s_i)$$

This is the point before cancellation where you would substitute the type-specific $f_{N,type(i)}(s_i)$ if you were to calculate without the λ_k^* simplification. If you proceed with the cancellation:

2.3. Combine and Cancel Common Product $P_0^*(D) = \prod_{i=1}^{M} f_{N,type(i)}(s_i)$ (Type-Specific f_N)

Define $\lambda_k^* = \frac{f_M(s_k)}{f_{N,type(k)}(s_k)}$. Any term $f_M(s_k)\prod_{i\neq k} f_{N,type(i)}(s_i)$ can be written as $\lambda_k^* P_0^*(D)$.

- $P(D \mid O) = \frac{1}{n}\sum_{j\in A}\lambda_j^* P_0^*(D)$
- $P(D \mid CF) = \frac{1}{n'_{phys}}\left(\sum_{j\in B_{act}}\lambda_j^* P_0^*(D) + (n'_{phys} - n^\dagger)P_0^*(D)\right)$
- $P(D \mid NF) = P_0^*(D)$

Substituting into $\frac{P(D|O)}{xP(D|CF)+(1-x)P(D|NF)}$:

$$\frac{P(O \mid D)}{P(N \mid D)} \;=\; \frac{\frac{1}{n}\sum_{j\in A}\lambda_j^* P_0^*(D)}{x\left[\frac{1}{n'_{phys}}\left(\sum_{j\in B_{act}}\lambda_j^* P_0^*(D) + (n'_{phys} - n^\dagger)P_0^*(D)\right)\right] + (1-x)P_0^*(D)}$$

$$= \frac{\frac{1}{n}\sum_{j\in A}\lambda_j^*}{x\left[\frac{1}{n'_{phys}}\left(\sum_{j\in B_{act}}\lambda_j^* + (n'_{phys} - n^\dagger)\right)\right] + (1-x)} \qquad \text{(Cancel } P_0^*(D) \text{ fr}$$

$$= \frac{\frac{1}{n}\sum_{j\in A}\lambda_j^*}{x\left(\frac{1}{n'_{phys}}\sum_{j\in B_{act}}\lambda_j^* + \frac{n'_{phys}-n^\dagger}{n'_{phys}}\right) + (1-x)} \qquad \text{(Distribute } 1/n'_{ph}$$

Final Exact Odds Formula (With Type-Specific f_N)

This is the rigorous expression, now explicitly accommodating type-specific non-match likelihoods $f_{N,type(k)}(s_k)$ within each λ_k^*.

$$\boxed{\frac{P(O \mid D)}{P(N \mid D)} \;=\; \frac{\frac{1}{n}\sum\limits_{k\in A}\frac{f_M(s_k)}{f_{N,type(k)}(s_k)}}{x\left(\frac{1}{n'_{phys}}\sum\limits_{k\in B_{act}}\frac{f_M(s_k)}{f_{N,type(k)}(s_k)} + \frac{n'_{phys}-n^\dagger}{n'_{phys}}\right) + (1-x)}}$$

- Each $f_{N,type(k)}(s_k)$ in the denominator of λ_k^* is specific to the type of non-matching trace k (and could be a mixture as per Steve's suggestion).

- This formula is "accurate" in the sense that it correctly applies Bayes' theorem given these more nuanced likelihood definitions. The challenge shifts to defining and computing each $f_{N,type(k)}(s_k)$.